Employing DRM Lists and GloVe Generated AI lists to Elucidate the Insensitivity of Recognition Tests for Detection of the Sleep Consolidation Effect in the Presence of Significant Confounding Effects Introduced by Differences across Test Delay and Learning Time for a 12-hour Retention Interval

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Abstract

Confounding sleep-consolidation effects stem from recent studies engaging a 12-hour retention interval and recognition-testing within a DRM-paradigm. We want to identify conditions that generate robust effect to eventually understand the intricate details of sleep-dependent memory-consolidations' plastic processes. We investigate potential explanations accounting for the conflicting literature. Due to the high-gist-based processing engaged, DRM lists may be inconvenient stimuli for sleep-dependent memory-consolidation studies. Accordingly, we manipulate List Type, Test-Delay, and Learning-Time between-subjects to compare True Recognition and False Recognition across critical conditions. Evening-acquisition relative to morning-acquisition, recognition tested at a 12-hr delay relative to immediate-testing and encoding AI-lists relative to DRM-lists yielded better True Recognition. Delayed-testing also yielded higher False-Recognition than Immediate-Testing. Importantly, no sleep-consolidation effect persists with significant differences across critical conditions due to the confounder variables despite encoding AI-Lists instead of DRM-Lists. These findings reinforce theoretical inferences that recognition-testing is largely insensitive to the sleep-consolidation effect as externally supplied cues likely covertly activate traces. Moreover, we also found that the relationship between List-Type and Learning-Time will mediate False Recognition. These findings highlight the problematic nature of comparing sleep-consolidation effects across studies due to differential semantic-properties of lists, retention-intervals, and roles of circadian-effects.

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The role of sleep in memory-consolidation is beyond dispute (Diekelmann & Born, 2010). An explicit function of sleep is challenging to derive owing to the multifaceted nature of the relationship linking sleep and memory. Generally, non-REM enhances verbatim-representations via the active context-dependent hippocampal-replay of episodic-traces, while overlapping gist-traces extracted overtime foster persistent, neocortical, abstract-representations during REM (Lewis et al., 2018). Moreover, Ellenbogen et al. 2006 illustrated that sleep protects verbatim-traces from subsequent associative-interference by demonstrating a 58% reduction in the cued-recall of A-B word-pairs for the wake (vs. sleep) condition after learning a competing interference list comprising of A-C word-pairs following a 12-hour retention interval. Current research focuses on identifying experimental conditions that unmask the sleep consolidation effect to understand the phenomena further (Diekelmann & Born, 2010). As recall-testing generates consistent effect sizes, this study is motivated by recognition-testing inconsistency (Newbury & Monaghan, 2018).

Using a DRM-paradigm Lo et al. 2014 showed sleep (vs. wake) reduced recollection-based false-recognition of critical-lures after an 8.5-hr retention-interval without affecting the veridical-recognition of studied words. Unlike Payne et al. 2009, this study did not incorporate a control-condition where participants undergo recognition-testing immediately following morning-acquisition and evening-acquisition. Thus, differences in learning and recognition performance due to circadian differences may have confounded the effect sizes yielded (Lo et al., 2014). Moreover, acquisition of DRM lists at 5:30 PM (vs. 8:30 AM) delivers better explicit recognition performance following a one-week delay (Barbosa & Albuquerque, 2008).

Additionally, improved source monitoring during the mornings may reduce false recognition due to enhanced recollection-based rejection (Newbury & Monaghan, 2019). Notably, when acquisition and recognition-testing took place during the evening for both sleep and wake conditions across a 44-hr retention-interval, neither true-recognition nor false-recognition differed across participants that stayed awake the first night (wake-condition) and those that slept (sleep-condition) (Diekelmann et al., 2008). Although Fenn et al. 2009 yielded similar results to Lo et al. 2014 employing a comparable design that incorporated immediate/control groups, the study did not account for baseline differences in performance across the 12-hr retention intervals independent of sleep consolidation (Lo et al., 2014). Multiple activations from repeatedly presenting DRM-lists strengthen episodic-representations to yield improved true-recognition (Dubuisson et al., 2012). Such activations may reflect time-dependent reorganizations within the hippocampus consistent across sleep/wake, which is consequently related to increases in false-recognition for explicit-memory tasks across delays due to increased reliance on gist-traces (Dandolo & Schwabe, 2018; Samantha et al., 2020; Flegal et al., 2010). Recently, Monaghan et al. 2017 used the DRM paradigm to show subsequent increases in false-recognition following a 12-hr retention-interval for sleep (vs. wake) conditions for words presented to the left-hemispheres. However, for evening-acquisition, similar lure-words accuracy across immediate/control and delayed/sleep-conditions suggest that baseline performance differences across sleep and wake conditions may have confounded the sleep-consolidation effect inferred (Monaghan et al., 2017).

Such inconsistencies may stem from the recruitment of more gist-based processing instead of explicit-encoding strategies due to DRM lists' highly semantic nature to yield immediate neocortical/abstract representations that do not benefit from sleep-consolidation (Pardilla-Delgado & Payne, 2017; Himmer et al., 2017). However, gloVe-generated wordlists may promote more verbatim representations as words are compiled based on co-occurrence in

the explicit environment, which is subsequently mediated by contextual-similarity (Pennington et al., 2014; Leach & Samuel, 2007). While Diekelmann et al. 2008 showed no sleep-consolidation effect for a 44-hr retention interval, this experiment aims to highlight the ineptitude of recognition-testing to capture sleep consolidation effects for a 12-hr retention interval. The study manipulates retention interval, learning time, and list type between subjects to demonstrate that a significant sleep consolidation effect will not persist after accounting for significant differences across critical conditions that stem from confounder variables despite employing a DRM-paradigm or a gloVe-paradigm.

We hypothesize participants who undergo evening-acquisition relative to morning-acquisition and delayed-testing relative to immediate-testing will yield higher true and false-recognition scores. Participants who encode AI-lists will yield higher true-recognition relative to DRM lists due to more explicit processing. Those that encode DRM lists will yield higher false-recognition due to more gist-based processing. No other differences across conditions, including sleep-consolidation effects, will reach significance due to recognition testing.

Methods

Participants

We recruited USYD undergraduates from a third-year psychology-cohort. The final data-set (n = 142) excluded incomplete data-sets and any submissions data-sets submitted outside the allocated 4-hr time-windows for both acquisition-phase and test-phase within a 12-hr-delayed-condition. Refer to Appendix for details.

Apparatus

Learning-stimuli comprised 120 Study-List-Items extracted from either eight DRM-lists (15/list) extracted from literature or eight gloVe-generated AI-lists (15/list). The recognition-test comprised 48/120 Study-List-Items for the eight respective DRM/AI-lists (6/list) and the corresponding 8 Unstudied-Lure-Items (1/list). Furthermore, it contained

48/120 Unstudied-Control-Target-Items from the eight respective control-lists (6/list), generated similar to DRM/AI-lists, and the corresponding 8 Unstudied-Control-Lure-Items (1/list).

Design

The study used a 2 x 2 x 2 between-subject factorial design. It manipulated List-Type (DRM-list vs. AI-list), Learning-Time (Evening vs. Morning), and Test-Delay (No-Delay vs. 12hr-Delay). The dependent variables comprised the True-Recognition-Rate/Corrected-Hit-Rate and the False-Recognition-Rate, reflective of Study-List-Item-Memory and False-Memory, respectively. It incorporated four experimental-conditions for each List-Type to study possible sleep-effects. Refer to Appendix for details.

Procedure

Learning-Phase

Participants either encoded 120 Study-List-Items from DRM-lists or AI-lists as they progressively appeared in the centre of their screens during the morning or evening.

Interference-Phase

Participants in the Immediate-Testing-Condition completed 24 math-questions. Participants in the 12hr-Delayed-Testing-Condition completed the same interference-task after a 12-hr retention-interval comprising sleep or wake.

Assessment-Phase

Participants then took a recognition-test requiring a 'OLD' or 'NEW' response to 112 items.

Statistical-Analyses

We used correct responses to calculate the Corrected-Hit-Rate/Corrected-True-Recognition-Rate and ratio of ‘OLD’ responses to calculate the Corrected-False-Recognition-Rate after accounting for Response-Bias. Post-hoc t-tests followed the two ANOVAs carried

out to assess main effects of all independent-variables' and sleep-effects from possible interaction-effects

Results

Refer to Figure 1. For Raw-Memory-Performance.

A significant main-effect was found for List-Type [F (1,134) = 8.52, p = .0041], Test-Delay [F (1,134) = 7.55, p =.0068], and Learning-Time [F (1,134) = 6.01, p = .0155] on the Corrected-True-Recognition-Rate and also Test-Delay on the Corrected-False-Recognition-Rate [F (1,134) = 4.54, p = .0349]. Refer to Figure 2 and 3 for comparison of effect-sizes for the means.

As shown in Figure 3, a significant interaction-effect was observed for List-Type and Learning-Time on the Corrected-False-Recognition-Rate [F (1,134) = 3.98, p = .0481].

Table 1 and 2 shows no other effects reached significance, $p > .05$.

Discussion

Expectedly, we did not find a sleep-consolidation effect. Moreover, evening-acquisition and AI-list-acquisition produced better true-recognition than morning-acquisition and DRM-list-acquisition, and delayed-testing yielded better true and false-recognition than immediate-testing. Unexpectedly, the relationship between stimuli learned (DRM/AI) and learning time (morning/evening) constrained false recognition.

Null effects of sleep on true-recognition is consistent with Lo et al. 2014, Fenn et al. 2009 and Diekelmann et al. 2008, who all employed recognition-testing with DRM-lists. Stepan et al. 2017 replicated this relationship using an eye-witness-memory-paradigm very distinct to the DRM-paradigm. Notably, these studies use neutral-stimuli, while studies generating a significant-effect engage emotional-stimuli. However, controlling for acquisition-memory-strength eliminates the robust sleep-benefit for emotional-stimuli relative to neutral-

stimuli (Schoch et al., 2017). Thus, this pattern likely reflects recognition memory-performance following sleep (Stepan et al., 2017).

Null effects of sleep on false-recognition is consistent with Diekelmann et al. 2008 who minimised circadian effects by equating learning and testing times across critical-conditions using a 44-hr retention interval. Comparatively, our study shows that a significant sleep-effect on false-recognition does not persist after accounting for confounding variables with a 12-hr retention interval. Such findings are inconsistent with Lo et al. 2014 and Fenn et al. 2009 who used similar retention intervals. The former did not incorporate control/immediate groups to account for circadian variations (Payne et al., 2009). The latter did not control for baseline-performance differences over retention intervals and response-bias across critical-conditions (Lo et al., 2014). Overestimated effect-sizes may follow interpretation of sleep-consolidation-effects from direct comparisons across critical-conditions without accounting for confounding-variables. Moreover, the sleep-condition may selectively benefit from better discriminability and source-monitoring processes for morning-recognition testing relative (Lo et al., 2014). In a similar study, Monaghan et al. 2017 showed no difference in false-recognition for evening/control groups relative to a sleep-condition despite showing an increase in the sleep-condition relative to the wake-condition. Notably, as the increase was specific for words displayed on the screen's right-side during recognition-testing, it is inconclusive if the behavioural-measure captured sleep's role in memory-consolidation or transferring lexical-information to the left-hemisphere (Newbery & Monaghan, 2018).

Despite manipulating list-type, an insignificant sleep-effect reinstates recognition-testing's insensitivity to sleep-benefits (Newbery & Monaghan, 2018). Largely, sleep's role is to facilitate accessibility via reactivating hippocampal-traces to promote integration into long-term neocortical-networks (Rasch & Born, 2013). Sensitive behavioural measures require subjects to re-activate traces via generating cues (Diekelmann & Born, 2010). Recall-testing

exploits this process to generate robust sleep-consolidation-effects (Payne et al., 2009). Contrastingly, insensitivity of recognition-testing follows covert-activation of traces via externally supplied cues (Diekelmann & Born, 2010).

The observed relationship between list-type and learning-time that constrains memory-performance is consistent with theories that implicate a change in learning-strategy from morning to night as the MTL's functional-connectivity shifts from displaying prominent local-connectivity to more frontoparietal-connectivity (Shannon et al., 2013). Moreover, mornings engage more explicit-strategies while afternoons facilitate more elaborative-strategies (Himmer et al., 2017). The latter may explain better overall memory performance for evening-acquisition (Oakhill, 1988). The fuzzy-trace-theory can account for better true-recognition for AI-lists than DRM-lists as the former may recruit more verbatim-traces while the latter may recruit more gist-traces (Brainerd & Reyna, 2002). Moreover, studies showing a selective increase in false-memories following sleep for 'low-performers' relative to 'high-performers' suggest the former may not benefit from the recollection-based-rejection supported by explicit-encoding due to higher reliance on gist-based-processing (Diekelmann et al., 2010). Increased true and false-recognition with delay is consistent with the activation-monitoring-framework where re-activation leads to consolidation and subsequent spreading-activation of critical-lures (Dubuisson et al., 2012).

The processing differences between our math-based interference-task and the learning-stimuli may account for the null sleep-effects observed by recognition-testing if the sleep-benefit for declarative memories is largely stabilisation against retroactive interference. Ellenbogan et al 2006 only unmasked a sleep-benefit for A-B word-pair recall after introducing associative-interference via learning A-C word-pairs before retrieval-testing. Moreover, as AI-lists and DRM-lists yielded similar false-recognition, both lists may be semantically similar enough to recruit comparable gist-based processing-strategies that generate very abstract-

representations that do not benefit from sleep (Pardilla-Delgado & Payne, 2017). These experimental-limitations prevent the conclusive determination of recognition-testings' insensitivity to the sleep-benefit. Repetition of the study should incorporate an interference-task that require subjects to study wordlists engaging processing that is similar to learning-stimuli akin to Ellenbogen et al. 2006. Repetitions should also modify AI-lists to selectively exclude semantically-similar words and include contextually-relevant words to increase explicit-processing to further determine the inconvenience of using DRM-lists in sleep-consolidation studies (Pardilla-Delgado & Payne, 2017).

Conclusively, this study touches on the confounding-variables that may account for the inconsistent sleep-consolidation effects generated by previous studies that engage 12-hr retention-intervals within the DRM-paradigm. It stresses for caution when it comes to comparisons across studies to make general inferences about the conditions that facilitate sleep-dependent memory-consolidation. Studies may comprise differences in learning stimuli, circadian effects of different learning times, and different retention intervals, all of which have the potential to generate confounding sleep consolidation effects. The insight provided on the experimental conditions that may/may not generate sleep-consolidation effects is relevant for future research into the plastic-processes associated with sleep-dependent memory-consolidation (Diekelmann & Born, 2010).

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Table 1.

ANOVA-Table for effects of List-Type, Test-Delay and Learning-Time on the True-Recognition-Rate

Source	Sum Sq.	d.f.	Mean Sq.	F	Prob > F
IsDRM	.39057	1	.39057	8.52	.0041
IsDelay	.34598	1	.34598	7.55	.0068
LearnEve	.27551	1	.27551	6.01	.0155
IsDRM∗IsDelay	.00386	1	.00386	0.08	.7721
IsDRM∗LearnEve	.14805	1	.14805	3.23	.0745
IsDelay∗LearnEve	.00966	1	.00966	0.21	.647
IsDRM*IsDelay*LearnEve	.00928	1	.00928	0.2	.6535
Error	6.14062	134	.04583		
Total	7.39538	141			

Note. ANOVA analysis results for the main effects and the possible interaction effects of Test-Delay, Learning-Time, and List-Type on the Corrected True Recognition Rate (Hit Rate). We collated 142 datasets after participants completed a recognition test comprising of either DRM lists or AI lists, during the morning or evening and immediately following the learning phase of the study or after a 12-hour delay (± 4 hours). We then generated a True Recognition Rate that was reflective of Study List Item Memory Performance for each condition using the ratio of correct responses to both OLD and NEW words. We corrected the values for Response Bias to account for baseline Hit Rates to yield the Corrected True Recognition Rate for respective conditions. We used these values to run an ANOVA analysis. We found a significant main effect for Test-Delay [F (1,134) = 7.55, p =.0068, Learning-Time [F (1,134) = 6.01, p = .0155] and List-Type [F (1,134) = 8.52, p = .0041] on the Corrected True Recognition Rates. These

identifications allowed for suitable post hoc analyses. We observed no other significant effects across conditions, including any effects generated due to sleep consolidation, $p > .05$.

Table 2.

ANOVA-Table for effects of List-Type, Test-Delay and Learning-Time on the False-Recognition-Rate

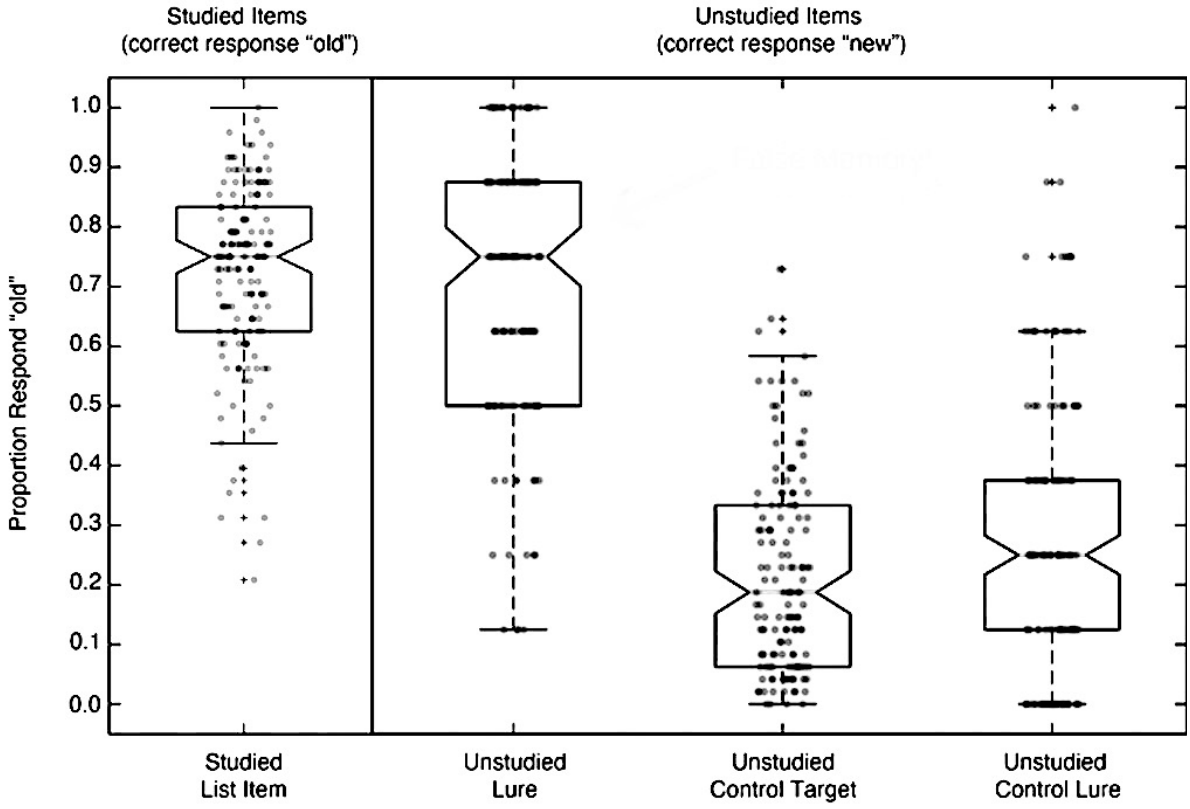
Source	Sum Sq.	d.f.	Mean Sq.	F	Prob > F
IsDRM	0	1	.00002	0	.9875
IsDelay	.3679	1	.36791	4.54	.0349
LearnEve	.0017	1	.00165	.02	.8866
IsDRM∗IsDelay	.0131	1	.01314	.016	.6877
IsDRM∗LearnEve	.3223	1	.32229	3.98	.0481
IsDelay∗LearnEve	.0961	1	.09608	1.19	.2781
IsDRM*IsDelay*LearnEve	.0041	1	.0041	0.05	.8224
Error	10.8536	134	.081		
Total	11.7039	141			

Note. ANOVA analysis results for the main effects and the possible interaction effects of Test-Delay, Learning-Time, and List-Type on the Corrected False Recognition Rate. We collated 142 datasets after participants completed a recognition test comprising of either DRM lists or AI lists, during the morning or evening and immediately following the learning phase of the study or after a 12-hour delay (± 4 hours). We then generated a False Recognition Rate that was reflective of False Memory for each condition using the ratio of 'OLD' responses to the Unstudied Lure Items. Although these items are related to lists learned at acquisition, they are new and not previously displayed during acquisition. We corrected the False Recognition Rates for Response Bias to account for the Baseline False Alarm Rates to yield the Corrected False Recognition Rate for the respective conditions. We used these values to run an ANOVA analysis. We found a significant effect of Test Delay on the Corrected False Recognition Rate

[F (1,134) = 4.54, p = .0349]. Also, we found a significant interaction effect for Learning Time and List Type on the Corrected False Recognition Rate [F (1,134) = 3.98, p = .0481]. These identifications allowed for suitable post hoc analyses. We observed no other significant effects across conditions, including any effects generated due to sleep consolidation, p > .05.

Figure 1.

Raw-Memory-Performance

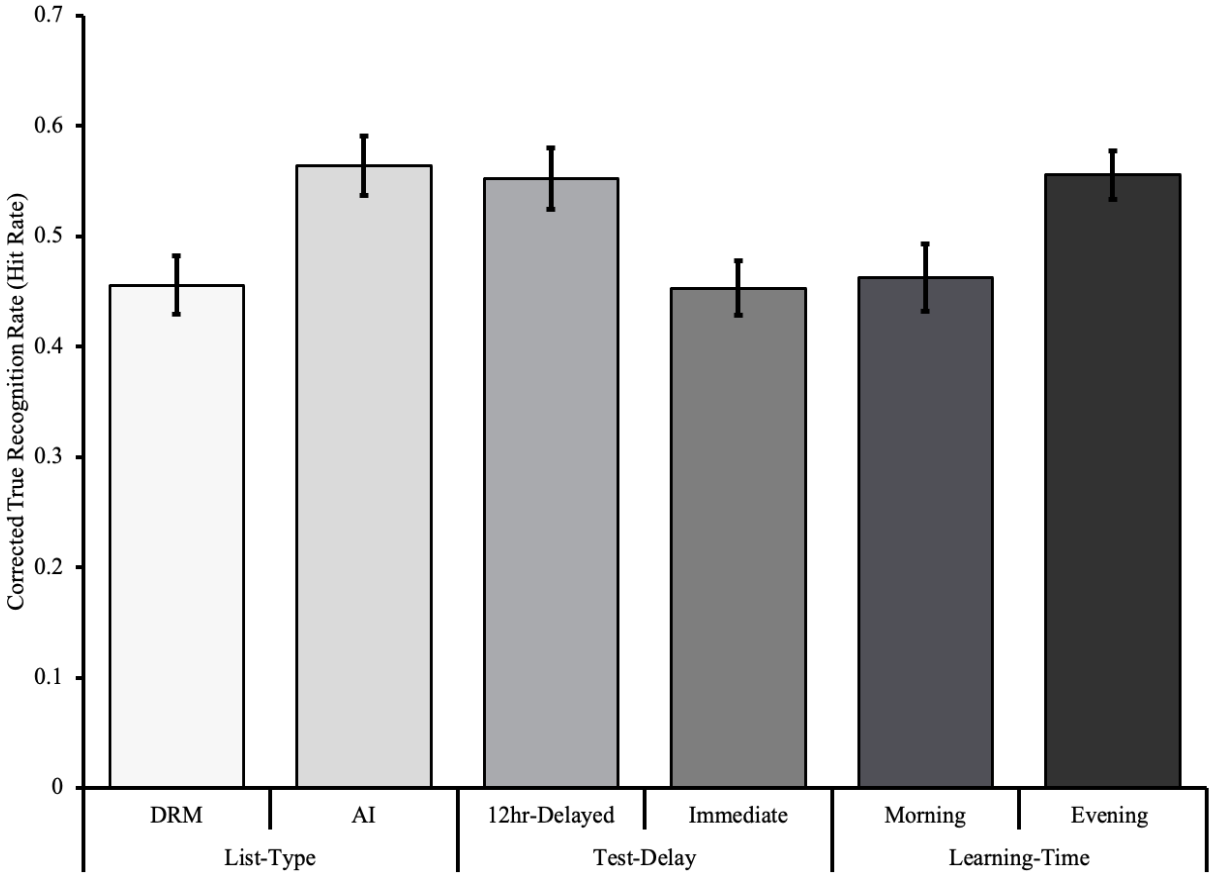


Note. This figure displays 142 raw data sets generated from the recognition tests administered after learning DRM lists or AI lists, during the evening or morning, and immediately following the experiment's acquisition phase or following a 12-hour delay (± 4 hours). We generated the data based on the proportion of 'OLD' responses to the 4 types of words displayed at recognition testing. Type 1 comprised 48 out of 120 Study List Items presented during the acquisition phase of the experiment. The acquisition phase required the participants to learnt 8 DRM lists or gloVe-generated AI lists. We adopted the DRM lists from existing literature. There was a total of eight lists that comprised of 15 words. The Study List Items during the recognition test only 6 of the 15 words used in each DRM list. We generated the AI list by inputting a specific critical word to generate 15 related words. The acquisition phase for participants encoding AI lists comprised eight sets of these 15 related words. The Study List Items for the recognition test for participants encoding AI lists comprised 6 of these 15 words

from each list. In both AI and DRM conditions, a OLD response to the 48 Study List Items was correct. The plotted data shows that participants reported correctly 73% of the time. Word type 2 comprised 8 Unstudied Lure Items. We investigated False Memory using this word type. Each of these words exhibited a relationship with the corresponding word lists studied during the experiment's acquisition phase. However, presentation of these words only took place during the recognition test. Thus, an OLD response for the Unstudied Lure Items was incorrect. A high proportion of incorrect responses to the Unstudied Lure Items displayed in the plotted data is reflective of False Memory. Specifically, participants incorrectly reported these words as NEW 72% of the time. Word Type 3 comprised of 48 Unstudied Control Targets. Like word type 1, we generated these words using gloVe or adopted them from existing DRM list literature. Presentation of these words only took place during recognition testing. Hence, an OLD response to this word type is incorrect. Since they displayed no relationship with the Study List Items, the plotted data shows a low proportion of incorrect responses. Specifically, participants incorrectly reported these words as OLD only 21% of the time. Finally, word type 4 comprised of 8 Unstudied Control Lures. These words displayed relationships with the corresponding eight control lists generated to yield the 48 Unstudied Control Targets. Like the Unstudied Control Targets, we restricted presentation of the 8 Unstudied Control Lures to the recognition test only. Thus, an OLD repose to this word type during the recognition test was incorrect. Moreover, since they exhibited no relationship with the Study List Items, the plotted data shows that participants made low proportions of incorrect responses. Specifically, participants only incorrectly reported these words as OLD 25% of the time. In each plot, that corresponds to one of the 4-word types, the individual dots reflect data collected from one specific person. Furthermore, for each plot, the central part of the box is the median, the edges of the box are the 25th and 75th percentiles, the whiskers extreme datapoints, and the outliers are plotted as individual crosses.

Figure 2.

Comparison of Main Effects of List-Type, Test-Delay and Learning-Time on the Corrected-True-Recognition-Rate

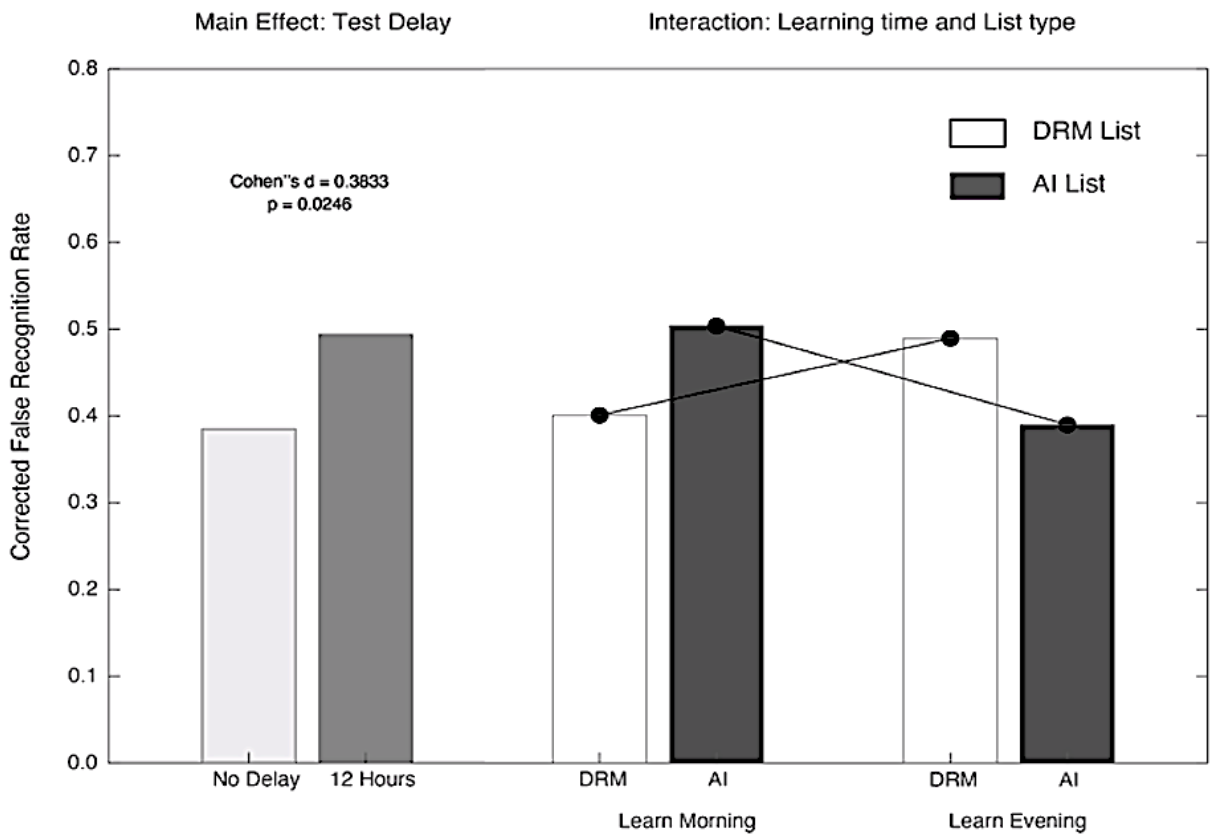


Note. This data plots the Corrected True Recognition Rates' effect sizes for the three main effects of List-Type, Test-Delay and Learning-Time. ANOVA analyses for 142 complete data sets submitted on time after a recognition test comprising of either DRM or AI lists, during the morning or evening, immediately following acquisition of respective wordlists or after a 12-hour retention interval, showed significant main effects of List-Type, Test-Delay and Learning-Time. We carried out post hoc unpaired t-tests to compare the mean Corrected True Recognition Rate for DRM vs. AI lists, 12-hr-Delayed-Reognition-Testing vs. Immediate/No-Delay-Recognition-Testing and Morning-Learning vs. Evening Learning. Participants who encoded the AI lists yielded a significantly higher mean Corrected True Recognition Rate/Hit

Rate (M = .564, SD = .0268) than those that encoded the DRM lists (M = .4558, SD = .0268); p = .0045, d = .4847. After a delay, participants who engaged in recognition testing yielded a significantly higher Corrected True Recognition Rate (M = .5524, SD = .0277) than those engaging in recognition testing immediately following the study's acquisition phase without a retention interval (M = .4538, SD = .0246) ; p = .0097, d = .4422. Furthermore, participants who encoded study lists in the evening yielded a significantly higher Corrected True Recognition Rate (M = .5556, SD = .0222) than those who encoded study lists during the morning (M = .4623, SD = .0306); p = .0148, d = .4144. The error bars are representative of the standard error of the means.

Figure 3.

Main Effect of Test-Delay and Interaction Effect of Learning Time and List Type on the Corrected-False-Recognition-Rate



Note. This data plots the Corrected False Recognition Rates' effect sizes for the main effect of Test-Delay and Interaction effect of List Type and Learning Time. ANOVA analyses for 142 complete data sets submitted on time after a recognition test comprising of either DRM or AI lists, during the morning or evening, immediately following acquisition of respective wordlists or after a 12-hour retention interval, showed a significant main effect of Test-Delay and a significant interaction effect of Learning-Time and List-Type. We then carried out post hoc t tests to compare the mean Corrected False Recognition Rate for the 12-hr-Delayed-Recognition-Testing condition to the Immediate-Recognition-Testing condition. Participants who engaged in recognition testing following a 12-hour retention interval yielded a significantly higher mean Corrected False Recognition Rate compared to those engaging in

retrieval testing immediately following the learning phase of the experiment without a retention interval; $p = .0246$, $d = .3833$. Data was unavailable for post hoc analyses of the significant interaction effect observed for List Type and Learning Time on the Correct False Recognition Rate. Thus, a directionality for this interaction effect is inconclusive.

Appendix

Each List Type (DRM/AI) had 4 experimental conditions. The following table presents number of participants assigned to each condition contributing to the final 142 data sets:

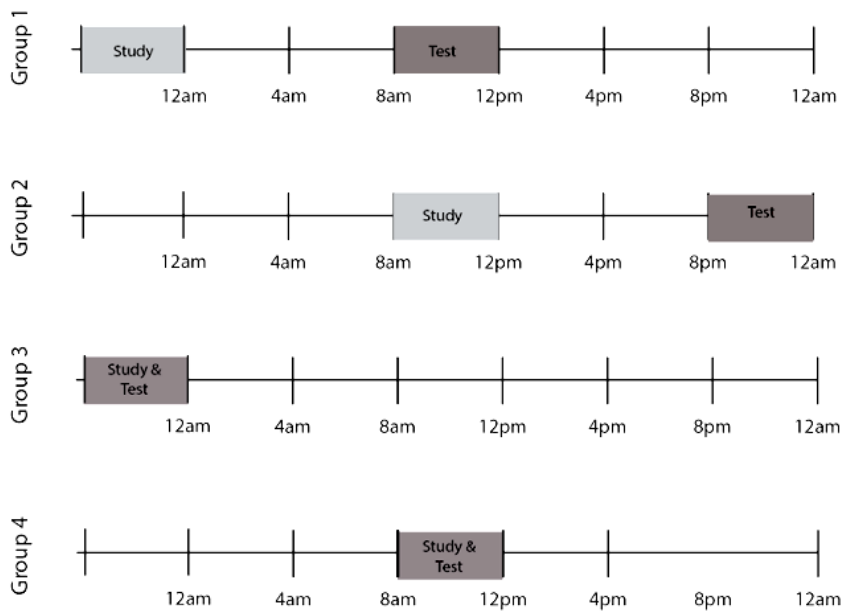
For DRM Lists:

	Morning Learning	Evening Learning
12-hr/Delayed	21	21
No-Delay/Immediate	18	14

For AI Lists:

	Morning Learning	Evening Learning
12-hr/Delayed	19	17
No-Delay/Immediate	17	15

The following timeline displays the allocated time windows for the respective conditions for different phases of the experiment.



The following table contains details of each of the specific experimental conditions used for the different lists.

Condition	Morning Learning	Evening Learning
12hr /Delayed	Condition 1 Participants studied their respective wordlists between 8 AM and 12 PM. They then returned after a retention interval of 8 to 16 hours comprising of wakefulness. Following the interference task, they engaged in a recognition test. This task had to be completed in between 8 PM and 12 AM.	Condition 2 Participants studied their respective wordlists between 8 PM and 12 AM. They then returned after a retention interval of 8 to 16 hours comprising of sleep. Following the interference task, they engaged in a recognition test. This task had to be completed in between 8 AM and 12 PM.
No-Delay /Immediate	Condition 3 Participants studied their respective wordlists between 8 AM and 12 PM. They then immediately engaged in an interference task. Following the interference task, they engaged in a recognition test. Participants were required to complete this task before 12 PM.	Condition 4 Participants studied their respective wordlists between 8 PM and 12 AM. They then immediately engaged in an interference task. Following the interference task, they engaged in a recognition test. Participants were required to complete this task before 12 AM.